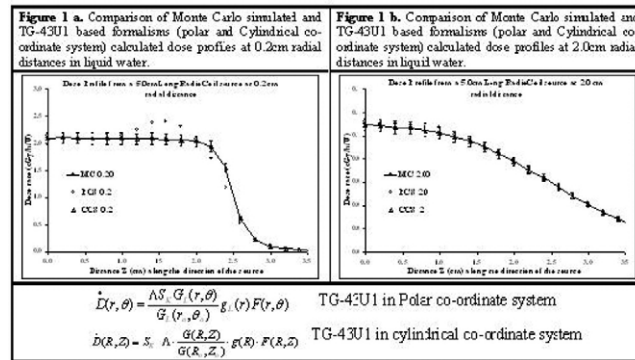




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2861 Ru-106 Eye Plaques for Treatment of Ocular Melanoma - Practical Issues

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Purpose/Objective(s): At our institution, Ru-106 plaques have been used in to complement traditional I-125 COMS plaques for radiotherapy management of uveal melanoma. Over 45 patients have been treated with Ru-106 since the launch of this program in late 2003. We report on our experience with treatment planning considerations and quality assurance of Ru-106 beta-emitting plaques.

Materials/Methods: Six sources (2 of each model: CCB, COB, and CCA, manufactured by BEBIG GmbH, Berlin, Germany) were commissioned since Dec. 2003. The CCB and CCA plaques are fully circular with a 20-mm and 15-mm diameter, respectively. The COB is 20 mm in diameter with a notch. Measurements of the absolute dose rate and relative dose uniformity were obtained using radiochromic films and a hemispherical eye phantom. A high resolution CCD densitometer (PeC 100, Photoelectron Co.) was used to digitize the films (0.13 mm pixel resolution). Sr-90/Y-90 source traceable to the National Institute of Standards and Technology (Gaithersburg, MD) was used for film calibration. A scaling function, for converting the measured dose rate in plastic to that in water, was estimated with Monte Carlo simulations (MCNPX code, Los Alamos National Lab., Los Alamos, NM). Our overall measurement uncertainty is $\pm 11\%$ (2σ). The measured dose distributions are used for plaque commissioning and treatment planning. Treatment planning was done using an Excel spreadsheet and a SURFER software program (Golden Software, Inc., Golden, CO) for isodose contour plotting.

Results: The absolute dose rate, along the central axis, for each source model is found to be in good agreement (within $\pm 10\%$) with the manufacturer's reported values. All 6 tested plaques were found to have good dose uniformity at measured a nominal depth of 2 mm from the inner surface of the plaque (within $\pm 10\%$). At our institution and over the last 3 years, the most commonly used plaque was the notched 20-mm COB ($n = 20$), followed by the circular 20-mm CCB ($n = 15$) and the 15-mm circular CCA ($n = 10$). Relative to I-125 COMS, our results indicate that Ru-106 delivers lower dose to surrounding structures due to its steep depth dose curve; while reasonable doses are delivered to the sclera if tumor apical height within 5 mm is indicated. For tumors close to the edge of the optical disk, we found the notched COB plaque to deliver lower dose to the optical disk than I-125.

Conclusions: We performed an independent verification of BEBIG Ru-106 plaques dosimetry. Implementation of this quality assurance program insures accurate Ru-106 radiotherapy treatment planning and delivery. We find Ru-106 plaque preparation logistics to be easier than I-125 COMS due to reusability over its one-year half-life and the allowable 50 sterilization cycles. Our surgeons also prefer Ru-106 for its thinner profile with easier insertions and less trauma to surrounding tissue.

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2862 Dosimetric Impact of Partial Gland Irradiation Utilizing Functional Image Guidance in Men Undergoing Brachytherapy for Localized Prostate Cancer

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Purpose/Objective(s): The objective of this simulation study was to provide proof of the dosimetric efficacy of using MRSI to guide partial gland irradiation in order to minimize the side effects and produce equivalent outcome.

Materials/Methods: Eight subjects with localized prostate cancer treated on prospective Phase II study utilizing Magnetic Resonance Spectroscopic Imaging (MRSI) to guide prostate brachytherapy (PB) were simulated in this dosimetric impact study. All patients had histological confirmed adenocarcinoma of the prostate, clinical stage T1c or T2a, PSA ≤ 10 , and Gleason score of 6 or less. Biologic tumor volumes (BTV) that were previously utilized in a Phase II prospective study were utilized for this dosimetric simulation study of partial gland irradiation. In this simulation study, only the BTV within the prostate gland was treated with I-125 radioactive seed implantation to a dose of 160 Gy.

Results: The simulated dosimetry and DVH were analyzed and compared to whole gland brachytherapy. We found the dose to both urethra and rectum much reduced. As shown in Table 1, the median D90 for urethra was only 23.6 Gy for partial gland irradiation, while 159.1 Gy for whole gland irradiation. The median V100 for rectum was 21.6% for whole gland irradiation and it decreased to 2.0% for partial gland irradiation.